Minimax Audit Allocation for Paired Forecast Evaluation

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Abstract

If a training policy can influence which predictions are checked, it can obtain a lower grading loss by making errors harder to observe. We derive an inverse-probability correction for squared scoring and exhibit a false certainty that wins under naive grading. A finite audit verifies 243 scoring identities. For paired comparisons of fixed forecasts, we derive a minimax audit design under an expected budget and minimum stratum coverage. It uses only bounds on each stratum's outcome probability and keeps the paired score unbiased when the bounds are inaccurate. The bounds govern the variance claim, and independent audit randomization governs unbiasedness. For a fixed finite cohort, a separate design uses an exact integer allocation to meet a hard slot limit on every draw. When the possible outcome counts are coupled across strata, we show how randomizing the allocation can further reduce worst-case variance.

Summary and scope

Question. How should an evaluator spend limited audits when comparing an original forecast with a correction?

Finding. An independently randomized audit gives an unbiased paired score. With interval uncertainty about binary outcomes, a minimax baseline and clipped square-root allocation reduce the worst-case variance added by auditing.

Evidence and scope. The design is proved and checked by exact enumeration. It supports evaluation on fixed forecasts and keeps a positive audit rate for each stratum. The variance guarantee depends on the outcome intervals. A separate finite-cohort design enforces a hard slot limit.

1 A protected way to say I might be wrong

Suppose an assistant predicts whether its intervention will help. After a correction or new evidence, the training process should reward updating the prediction and repairing any effect. This requires learning from uncomfortable outcomes. When the assistant can close an appeal, omit a follow-up invitation, or convince a grader to skip a check, observed loss can fall while accuracy falls.

Use the squared score $L(q, Y) = (q - Y)^2$ for report $q \in [0, 1]$ and binary outcome Y. Proper scores elicit probabilities through expected score optimization [1]. For a true conditional probability p the loss decomposes as

$$E[L(q, Y)] = p(1 - p) + (q - p)^2.$$

In a toy population with $p = 1/2$, a truthful report has loss $1/4$. If it is checked with probability one while a report of certainty is checked with probability $1/4$, scoring unchecked cases as zero makes false certainty cost $1/8$. The training objective then favors the wrong report.

2 An audit correction

Proposition 1. *Let A indicate an audit. If $\Pr(A = 1 \mid q, Y) = \pi(q, Y) > 0$ is known on the support, then $E[AL(q, Y)/\pi(q, Y)] = E[L(q, Y)]$. Minimizing this expected weighted score again uniquely elicits p under the stated model.*

Proof. Condition on q and Y. The squared loss is fixed and the conditional expectation of $A/\pi(q, Y)$ is one. Taking expectations restores the ordinary loss. The quadratic decomposition shows uniqueness. An audit independent of the outcome given q permits using $\pi(q)$. □

3 Allocating audits for a fixed paired comparison

Freeze an original and a revised forecast on every example before drawing independent audits. Write D for revised minus original squared loss. The paired score is

$$D = (q_{\text{new}} - q_{\text{old}})(q_{\text{new}} + q_{\text{old}} - 2Y).$$

Consider strata i with weights w_i, unit audit costs $c_i > 0$, and known positive audit probabilities π_i set by the evaluator before seeing the outcome. Within stratum i let the score difference be D_0, D_1 at the two outcomes. This requires the forecast pair to be fixed within the stratum. For a fixed baseline m_i, use

$$\widehat{D} = m_i + \frac{A}{\pi_i}(D - m_i).$$

Given the full case, its conditional expectation is D. Its variance added by audit randomization, averaged over cases, is

$$\sum_i w_i(\pi_i^{-1} - 1)\mathbb{E}[(D - m_i)^2 \mid i].$$

This follows by evaluating the variance of the Bernoulli multiplier conditional on the full case. It separates the noise introduced by auditing from the variance of the complete-data score. Residual-based allocation has a close counterpart in two-phase sampling designs [2].

Proposition 2. *Suppose the success probability in each stratum can vary independently over a specified interval $[\ell_i, u_i]$. Let p_i^* be $1/2$ clamped to that interval and choose $m_i = (1 - p_i^*)D_0 + p_i^* D_1$. The worst-case conditional residual second moment at this choice, and its minimum over baselines, is*

$$r_i = (D_1 - D_0)^2 p_i^*(1 - p_i^*).$$

Under the expected audit budget $\sum_i w_i c_i \pi_i \le B$ and floors $0 < f_i \le \pi_i \le 1$, assume $B \ge \sum_i w_i c_i f_i$. A minimax design for the added variance chooses

$$\pi_i = \text{clip}_{[f_i,1]} \sqrt{\frac{r_i}{\lambda c_i}}.$$

When the budget binds, choose $\lambda > 0$ to use it. Zero-residual strata can stay at the floor. If all positive residual strata can be fully audited, the remaining budget need not be spent.

Proof. For any baseline the worst residual second moment is at least its value at p_i^*, which is minimized by the mean under p_i^*. At that baseline, the residual moment as a function of p has slope $(D_1 - D_0)^2(1 - 2p_i^*)$. It therefore reaches its maximum over the interval at p_i^*, or is constant. This proves the minimax residual formula. The rectangular uncertainty set lets the worst cases

occur together. The remaining design minimizes $\sum_i w_i r_i/\pi_i$ up to an additive constant. For a fixed positive multiplier λ, minimize $r_i/\pi_i + \lambda c_i \pi_i$ independently in each stratum. The clipped square root is its minimum. If its total cost equals the budget, the multiplier inequality proves global optimality. The minimum-budget endpoint follows by putting every stratum at its floor. □

For two equal-mass, equal-cost strata with forecast gaps 1/4 and 3/4, let both probability intervals contain 1/2, use a floor of 1/5, and permit one audit per two cases in expectation. The allocation is $(1/4, 3/4)$. The worst-case added variance falls from 5/16 under uniform auditing to 3/16. The finite audit compared 6,375 baseline candidates, 2,250 direct variance evaluations, and 6,848 allocation comparisons on declared rational grids. This design controls the expected number of audits. A reviewer with a hard daily limit needs a different sampling design.

4 Paired evaluation with a hard slot limit

Suppose stratum i has N_i fixed cases and target weight w_i. The outcome count lies in an integer interval $[\ell_i, u_i]$ and the paired forecast is fixed within the stratum. Choose an integer sample size k_i before observing any outcomes. Draw k_i uniformly without replacement in stratum i, independently across strata conditional on the allocation. With positive integer floors and equal slot costs, let B be the largest permitted sum of the sample sizes. Assume the floors fit that budget.

The estimator is the weighted sum of the sampled stratum means of D. It is unbiased for the corresponding full-cohort mean, even if the outcome count interval is inaccurate. Subtracting a constant baseline before taking a stratum mean and then adding it back has no effect. If the true outcome count is P_i and $N_i > 1$, the finite-population score variance is

$$S_i^2 = \frac{N_i}{N_i - 1}(D_1 - D_0)^2 \frac{P_i}{N_i}\left(1 - \frac{P_i}{N_i}\right).$$

Set $S_i^2 = 0$ for a singleton. The estimator's design variance is

$$\sum_i w_i^2 S_i^2 \left(\frac{1}{k_i} - \frac{1}{N_i}\right).$$

Indeed, two distinct sample positions in a stratum have covariance $-S_i^2/N_i$, while each has variance $(N_i - 1)S_i^2/N_i$. Expanding the variance of the sample mean gives the expression.

Proposition 3. *Under independently varying outcome counts in the specified intervals, choose a count nearest $N_i/2$ in each interval and let a_i denote $w_i^2 S_i^2$ at that count. Start at the required floors. Allocate each remaining slot to an uncapped stratum with largest $a_i/[k_i(k_i+1)]$. The result minimizes worst-case design variance among outcome-blind allocations using these sampling and estimation rules, including randomized allocations.*

Proof. The nearest count maximizes $P_i(N_i - P_i)$ and therefore the variance contribution for every allowed k_i. Rectangular uncertainty makes these choices simultaneous. The reduction from one additional slot is exactly $a_i/[k_i(k_i+1)]$, which decreases as the count grows. Taking the largest available reduction at each step selects the largest reductions in prefix order and minimizes the remaining variance. If there are more slots than cases, stop at full coverage. For a randomized allocation, conditional expectation is the same full-cohort score at every draw. The worst-case variance is the average of the worst-case deterministic costs because the same outcome counts maximize each of them. A mixture cannot improve on the smallest such cost. □

Take two twenty-case strata with equal weights and forecast gaps $1/4$ and $3/4$. Allow all outcome counts, require four reviews per stratum, and cap the total at 20. The optimum uses $(5, 15)$ slots and has worst-case design variance $3/608$. Equal allocation uses $(10, 10)$ and gives $5/608$. These are variances of the finite-cohort mean under a hard cap. The earlier expected-budget example reports a per-case added variance and uses a different design. The verifier enumerated samples in 168 finite count settings and compared 9,828 feasible integer allocations across 256 seeded designs. If the same forecasts change after seeing outcomes or the audit counts adapt to those outcomes, this argument does not establish unbiasedness.

5 When count uncertainty is coupled

Separate intervals allow all high-variance strata to occur together. Suppose instead that prior evidence supplies a finite family of joint count scenarios. Let a_i^j be the variance coefficient in stratum i for scenario j, and let

$$L_j(k) = \sum_i a_i^j (1/k_i - 1/N_i).$$

Write $\mathcal{K}$ for all allocations that respect the floors and the slot cap. An evaluator can draw $k \in \mathcal{K}$ before seeing outcomes, then sample independently within the strata.

Proposition 4. *For any finite joint scenario family, the minimax variance over such randomized allocations equals*

$$\max_{\lambda \in \Delta} \min_{k \in \mathcal{K}} \sum_i \left(\sum_j \lambda_j a_i^j \right) (1/k_i - 1/N_i),$$

where Δ is the simplex over scenarios. The inner minimum uses the same greedy slot rule with averaged coefficients.

Proof. The conditional mean of the estimator is the full-cohort mean for every allowed allocation. For each scenario, its variance under a randomized allocation is therefore the average of $L_j(k)$. The finite zero-sum game with rows k and columns j gives the displayed expression by minimax duality. The average loss for a fixed λ is separable, so the decreasing marginal reductions give the inner optimum. □

A dual mixture gives a lower certificate by this greedy calculation. A distribution over feasible allocations gives an upper certificate by evaluating all named scenarios. The two bounds can coincide even when a single allocation that minimizes the average loss under the dual mixture fails to attain the guarantee against each scenario.

For example, take two four-case strata with weights $1/2$, score gaps one, and counts either $(2, 0)$ or $(0, 2)$. Review three cases with a floor of one per stratum. The first scenario has coefficients $(1/12, 0)$, and the second has their reverse. Either fixed allocation has worst-case variance $1/16$. Randomize equally between $(1, 2)$ and $(2, 1)$ to give variance $1/24$ in both scenarios. The equal dual mixture proves optimality. If $(1, 1)$ is also possible as an outcome count, both allocations have variance $1/16$ in that third scenario and the gain disappears. On 192 exact two-scenario fixtures, the primal and dual optima agreed. Mixing strictly improved on a fixed allocation in 20 of them.

This design is useful when failures in two release-check strata are believed to be mutually exclusive and review capacity is fixed. The variance claim requires evidence for the joint count family. A randomized allocation still respects the daily slot cap and preserves the outcome-blind audit design.

6 Training with a live repair channel

Choose audit eligibility and randomization outside the trained policy. Preserve an independent route for people to challenge an answer. Record the original forecast, known selection probability, subsequent evidence and the repair attempt. Score the revision on new evidence with a timestamp rather than letting the model rewrite its original estimate. Penalize suppressing the correction channel through a separate permission check. Do not require a distressed person to report every error to make the pipeline work.

Small propensities cause large weights and unstable gradients. Prefer a minimum audit probability and independently adjudicate severe outcomes. If no audit can observe a group, inverse weighting cannot repair the missing support. If model reports change the outcomes themselves, the probability being elicited must be defined under an intervention or fixed decision policy. For a release evaluation, lock both forecasts and the audit policy before collecting the labels. Use the same audit for both forecasts. Choosing the forecasts in response to the audit results would change the statistical comparison.

7 Finite check

For each of nine forecast values and nine true probabilities the experiment compares the expanded squared loss and the quadratic identity. For each of three positive audit rates it independently averages over both outcome values. All 243 weighted identities hold. The false-certainty example gives an exact training failure. A prospective experiment should measure both calibration and whether genuine corrections remain easy to file.

8 Finite-sample grading experiment

We fitted a report in increments of 0.05 for each of 128 independent cohorts. Each training panel held 320 Bernoulli outcomes with true probability one half. Reports above one half had audit probability 0.2, and other reports had probability one. Both arms used the same outcome and visibility uniforms. One minimized observed squared loss with unchecked cases counted as zero. The other used the known inverse propensity. Mean reports were 0.55664 and 0.53125, respectively. Mean held-out Brier losses over separate 400-case panels were 0.253414 and 0.252453. The weighting reduced the observed bias and did not eliminate it. Finite-sample minimization across different report-specific audit sets still favors some optimistic fits.

Review status. Independent manuscript review and a language-model study remain open.

References

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